Autonomous AI Civilization

Master PRD, Product Vision, Ten-Agent Society, Persistent Memory System, Child Synthesis Engine, UI/UX, Low-Level Architecture, Data Model, API Design, Workflows, Free-Tier Strategy, Pros, Cons, Testing, and Hackathon Roadmap

Core idea: A self-running observable world where nine specialized adults live persistent lives and one curious child learns from their distributed experiences, connecting memories across professions to create new, validated discoveries.

Prepared as a comprehensive one-shot project specification for hackathon planning and implementation.

Technology Summary

Layer	Recommended Technology	Primary Responsibility
Frontend	React + Vite + TypeScript	World map, agent follow mode, inspectors, timeline, discovery graph
Backend	Python + FastAPI	Simulation API, world engine, scheduling, validation, realtime events
AI Orchestration	LangChain	Prompt chains, context composition, structured outputs
Reasoning Model	Grok API	Agent decisions, conversations, reflection, hypotheses, validation
Long-term Memory	Supermemory	Semantic storage and retrieval of agent experiences
Structured State	Supabase PostgreSQL	Agents, schedules, world truth, events, provenance, projects
Realtime	WebSocket or SSE	Push world events and agent updates to observer

Table of Contents

Executive Summary

This document specifies an autonomous AI civilization designed as a persistent, observable world rather than a conventional chatbot or user-controlled game. Approximately ten AI agents live inside a small settlement. They wake, travel, work, solve practical problems, form relationships, return home, exchange experiences, read, reflect, sleep, and continue into the next simulated day. Their lives are synchronized by a world clock and governed by deterministic rules, but meaningful decisions and conversations are generated through a shared Grok API orchestrated by a Python and FastAPI backend. Long-term experiential memory is stored through Supermemory, while structured world truth, relationships, schedules, provenance, resources, and simulation state are persisted in Supabase.

The defining mechanic is that agents are not omniscient. Each agent knows only what they directly observe, what another agent tells them, what they read, what becomes public knowledge, and what they validly infer. A farmer can experience drought without the engineer knowing about it. The farmer may return home and describe the problem. A child may later hear the story, visit the engineer, learn how water can be moved through pipes or channels, and connect those separate memories into an irrigation hypothesis. That hypothesis must then be validated against expertise, resources, and world rules before becoming a real project. This causal chain makes memory visible and consequential.

The recommended society contains nine specialized adults and one unusual child agent. The adults acquire deep but narrow experiences through farming, engineering, medicine, teaching, logistics, cooking, art, leadership, and natural research. The child has a flexible routine and unusually broad social access. It attends lessons, visits workplaces, listens to family conversations, reads books, asks questions, and performs nightly reflection. Its growth comes from cross-domain synthesis: connecting memories originating from different people and domains. The child is not declared superintelligent by prompt. It earns broader capability through accumulated experience, retrieval, questioning, testing, and validated discovery.

The user is primarily an observer. The main interface resembles a living game world populated by AI non-player characters. A top bar displays Day 0 or the current day number, simulated date, 24-hour clock, weather, and simulation speed. The central view shows a stylized settlement map with moving agent avatars. The observer can click any agent, enter follow mode, inspect current activity, location, mood, goals, relationships, schedule, recent memories, and meaningful experiences, then switch instantly to another agent. A timeline records important events. A discovery-provenance view shows how an original event became a memory, moved socially, combined with another experience, produced a hypothesis, and changed the civilization.

For a hackathon, the complete vision must be narrowed into a reliable vertical slice. The strongest demonstration is one accelerated multi-day scenario with ten visible agents, deterministic routines, a seeded environmental problem, persistent individual memories, evening memory exchange, child cross-domain retrieval, one hypothesis, expert validation, and one visible civilization-level consequence. This proves the architecture without requiring a complete economic, political, generational, or biological simulation.

1. Product Vision, Thesis, and Positioning

The product vision is a world that remains interesting even when the observer does nothing. At 06:30 an agent wakes. At 07:00 a household eats breakfast. At 08:00 workers leave home. During the day the farmer inspects crops, the doctor treats illness, the engineer repairs tools, the merchant notices shortages, the teacher preserves knowledge, and the child explores. Events become experiences. Experiences become memories. Memories influence future decisions and conversations. Some memories travel to other people. Some become books or public records. Some combine unexpectedly and create discoveries.

The project should be described as an autonomous artificial society, a persistent multi-agent memory laboratory, and an observable emergent-world simulation. It should not be pitched merely as ten chatbots talking to each other. Random dialogue is easy to generate and difficult to care about. The product becomes meaningful when every conversation has context, every agent has information boundaries, actions alter persistent state, and later behavior can be traced to earlier experiences.

The strongest thesis is that memory creates continuity, and socially transmitted memory creates civilization. Without persistence, each agent is an isolated prompt. With private episodic memory, an agent develops a personal history. With relationships and conversation, experience moves through a social network. With books and public records, knowledge outlives the original event. With cross-domain synthesis, separate experiences can create innovation. The world therefore demonstrates several layers of intelligence rather than a single chat transcript.

The observer role is intentionally passive in the primary experience. Many AI products ask the user to continuously type instructions. Here, the world has agency of its own. The observer watches, follows, inspects, rewinds event history, and studies causal chains. Optional sandbox controls may later allow scenario creation, but the default hackathon experience should emphasize that the civilization moves forward automatically.

The project has a real-world architectural analogy. A company can be modeled as specialized agents representing engineering, operations, support, finance, sales, research, legal, design, and leadership. Each department knows different things. A cross-functional synthesizer can connect memories and identify opportunities. The village is an emotionally understandable visual metaphor for persistent organizational intelligence.

1.1 The One-Sentence Pitch

Ten autonomous AI agents live persistent lives in a small world, and one curious child learns from the specialized memories of everyone around them, connecting experiences across professions to create new, validated discoveries.

1.2 The Judge-Facing Answer to “Why Build This?”

Because most AI demonstrations reset, centralize knowledge, or hide memory behind a chat interface. This project makes long-term memory visible as a force that changes behavior, relationships, knowledge transfer, and innovation. It demonstrates how specialized autonomous agents can preserve individual experience, share it selectively, and create new value when memories from different domains are connected.

The simulation is not claimed to predict real human civilization. It is an experimental artificial-agent environment and a visual demonstration of persistent multi-agent memory, provenance, bounded autonomy, and long-horizon coordination.

2. Product Requirements Document

The product requirement is to create a persistent autonomous world that can run without direct user interaction, remain coherent across simulated days, expose agent behavior visually, and demonstrate memory-driven change. The system must balance three properties: autonomy, causality, and observability. Autonomy means agents continue living without user prompts. Causality means actions follow world rules and previous experiences. Observability means the user can understand what is happening and why.

The MVP audience is hackathon judges, AI developers, researchers interested in agent memory, and general users attracted by living-world simulations. The first session should require almost no explanation: the user opens the site, sees a world clock and moving agents, clicks a person, and immediately understands that the person has a role, routine, memories, relationships, and current activity.

The primary success criterion is not the number of generated messages. It is whether the product can visibly demonstrate a causal sequence in which an experience is stored, transmitted, retrieved later, combined with another experience, and used to produce a world-changing result. Secondary criteria include world coherence, visual clarity, agent individuality, API efficiency, and demo reliability.

The system shall support approximately ten persistent agents, a 24-hour simulated clock, day count, calendar date, adjustable simulation speed, deterministic routine scheduling, agent movement, locations, professions, households, relationships, individual memory namespaces, social memory transmission, books or public knowledge, AI-generated bounded decisions, structured actions, event logging, realtime frontend updates, and observer follow mode.

The system shall prevent arbitrary omniscience. Memory retrieval must be scoped. The system shall prevent the LLM from directly mutating world truth. Proposed actions must pass validation. The system shall preserve provenance for important memories and discoveries. The system shall continue routine simulation when noncritical AI calls fail.

The product should support a resettable demo seed. A live hackathon demo cannot depend entirely on unpredictable emergence. The environment may seed a drought, broken tool, shortage, or illness at a known time. The AI still reasons genuinely from supplied context, but the scenario guarantees that the system has something meaningful to reason about.

2.1 Functional Requirements

FR-1: The world clock advances according to configured simulation speed. FR-2: Each agent has a persistent identity, role, schedule, relationships, location, state, goals, and memory ownership. FR-3: Routine actions execute deterministically. FR-4: Meaningful decisions can invoke Grok through LangChain. FR-5: Agent outputs use validated structured schemas. FR-6: Direct observations can become episodic memories. FR-7: Conversations can create derived memories for listeners with provenance. FR-8: Relevant memories can be retrieved semantically. FR-9: The child can synthesize cross-domain hypotheses. FR-10: Hypotheses require validation before changing world truth. FR-11: The UI streams live events and supports agent follow mode. FR-12: The UI shows memory and discovery provenance. FR-13: Admin/demo controls support pause, resume, speed change, and reset.

2.2 Non-Functional Requirements

The application should remain responsive while AI calls are in progress. Realtime updates should arrive quickly enough to feel alive. The simulation must tolerate transient model failures. Secrets must remain server-side. Memory ownership must be enforced. Logs must make AI decisions traceable. The architecture must remain feasible on free or low-cost tiers and avoid unnecessary distributed infrastructure during the hackathon.

2.3 Out of Scope for MVP

Birth, death, genetics, unrestricted reproduction, hundreds of agents, warfare, complex politics, a full market economy, procedural terrain, unrestricted crafting, real scientific simulation, photorealistic 3D graphics, continuous voice acting for every agent, and fully open-ended user intervention should remain outside the first hackathon build.

3. Civilization Design: The Ten Agents

Ten agents provide enough diversity for specialization and social structure while remaining understandable. More agents multiply token cost, latency, UI clutter, relationship edges, and debugging complexity. The recommended roster uses nine adults with complementary domains and one child who functions as a cross-domain learner.

Every agent needs immutable identity fields and mutable life state. Immutable identity includes name, age category, profession, baseline personality, capabilities, family role, and communication style. Mutable state includes location, activity, mood, energy, needs, current goals, schedule exceptions, relationships, memories, beliefs, hypotheses, and recent events.

Agents should not be caricatures. The farmer is not only “the farming bot.” She can have family concerns, curiosity, preferences, and social history, but her profession gives her deeper exposure and competence in agriculture. This creates specialization without flattening personality.

3.1 Mira — Farmer

Mira manages crops, soil, seeds, irrigation status, pests, harvest, and weather observations. She wakes early, checks forecasts and field state, works at the farm, and returns home with experiences tied to ecology and food security. Her high-value memories include crop failure, unusual weather, successful planting methods, pest patterns, labor shortages, and resource constraints. She is likely to detect drought before most of the civilization.

Mira cannot create engineering solutions merely because the model knows about them. She may know simple agricultural techniques established in public knowledge, but specialized infrastructure requires communication with the engineer or access to appropriate books. This boundary is essential to the discovery narrative.

3.2 Arun — Engineer and Builder

Arun repairs tools, maintains structures, evaluates materials, builds approved projects, and understands mechanical systems. His routine includes workshop time, inspections, repairs, project work, and occasional consultations. He may know how water flows through channels or pipes without knowing that the north field is failing unless that information reaches him.

When another agent proposes a technical idea, Arun acts as a domain validator. He checks feasibility against available materials, labor, technology level, and known principles. He does not magically build instantly; validated projects create tasks and consume resources over time.

3.3 Sana — Doctor and Healer

Sana observes health, injuries, hygiene, nutrition, and disease patterns. Her memories can create important cross-domain links with water quality, food, housing, weather, and work conditions. She provides a second strong discovery path beyond irrigation: contaminated water may connect medical observations with engineering and public infrastructure.

3.4 Dev — Teacher and Librarian

Dev teaches foundational concepts, curates books, records validated discoveries, and helps convert personal experience into institutional memory. The library provides knowledge that agents did not personally experience. Reading creates source-tagged memories rather than omniscient knowledge injection.

Dev is central to cultural persistence. Once a discovery is validated and documented, future agents can learn it through public knowledge. This models the transition from episodic experience to civilization-level knowledge.

3.5 Leena — Merchant and Logistics Coordinator

Leena tracks inventories, shortages, deliveries, trade, and demand. She sees patterns that domain workers may miss: medicine consumption rising, food stock declining, wood becoming scarce, or a construction project creating pressure on supplies. Her experience can connect economics to engineering, agriculture, and health.

3.6 Kabir — Cook and Food Steward

Kabir manages meals and observes the practical consequences of agricultural output. The kitchen is also a social hub. Because many agents eat there or interact around food, Kabir naturally hears information from multiple domains. His role demonstrates that knowledge bridges need not always be formal experts.

3.7 Isha — Artist and Storyteller

Isha transforms events into stories, art, symbols, and cultural memory. She gives the world emotional texture and demonstrates that civilization is more than optimization. A major event may become a story that persists after precise factual details fade. Her output can influence morale, relationships, and public remembrance.

3.8 Rohan — Leader and Coordinator

Rohan receives reports, resolves priorities, allocates shared labor, and approves settlement projects. He is explicitly not omniscient. If no one reports the drought, he does not know it. Leadership quality therefore depends on information flow. This creates a realistic reason for meetings, reports, and public notices.

3.9 Nila — Researcher and Naturalist

Nila studies plants, weather, materials, and recurring patterns. She maintains hypotheses and performs bounded experiments. Her role helps distinguish observation from established knowledge. A model-generated idea remains uncertain until evidence or expert validation supports it.

3.10 Aadi — Child and Knowledge Synthesizer

Aadi is the central innovation. Unlike adults with long work blocks, the child has a flexible routine: lessons, play, workplace visits, reading, family meals, questions, and reflection. Aadi's advantage is breadth, not magical intelligence. The child meets people from different domains and can retrieve semantically related memories across those experiences.

The child should begin with limited knowledge and grow through lived experience. Metrics can include unique concepts encountered, trusted relationships, cross-domain connections, validated hypotheses, books read, unresolved questions, and discovery impact. The most important metric is not raw memory count but useful synthesis.

4. Daily Life, Fixed Routines, and Dynamic Exceptions

A world of completely random agents feels incoherent; a world of completely fixed schedules feels mechanical. The recommended model is bounded routine plus meaningful exceptions. Each agent has baseline routine blocks, and a scheduler advances them automatically. Weather, emergencies, invitations, illness, resource shortages, discoveries, unfinished goals, or relationship commitments may interrupt the routine.

A typical adult day contains waking, hygiene or preparation, breakfast, commute, morning work, lunch, afternoon work, return travel, dinner, social conversation, reading or leisure, reflection, and sleep. The exact times differ by profession. The doctor may be summoned. The farmer starts early. The teacher has lesson periods. The child has more exploratory freedom.

Routine transitions do not require Grok calls. If the clock reaches 08:00 and the farmer's next scheduled block is FARM_WORK, application code changes destination, emits movement events, and begins the activity. The LLM is invoked only when a meaningful decision exists. This is the foundation of free-tier feasibility.

At dinner, agents do not dump every memory. A salience selector chooses a small number of experiences based on importance, emotional weight, listener relevance, secrecy, trust, unresolved goals, and novelty. This produces natural memory exchange and limits token usage.

4.1 Example Farmer Schedule

05:45 wake; 06:15 breakfast; 06:45 walk to farm; 07:00 inspect fields; 08:00 planned work; 12:00 lunch; 13:00 afternoon work; 16:30 review crop state; 17:00 return home; 18:30 dinner; 19:00 family conversation; 20:00 reading, repair, or reflection; 21:30 sleep. Weather and emergencies can override blocks.

4.2 Example Child Schedule

06:45 wake; 07:15 breakfast; 08:30 lesson with Dev; 10:00 free exploration; 12:30 lunch; 13:30 workplace visit selected from valid destinations; 16:00 play or reading; 18:30 dinner; 19:15 conversation; 20:00 memory reflection and question formation; 21:00 sleep.

4.3 Exception Priority

Emergency health event outranks ordinary work. Severe weather may cancel travel. A critical resource shortage can trigger a leadership meeting. A promised social meeting may override optional leisure. The scheduler should encode priority rules so the LLM does not decide every conflict from scratch.

5. Autonomous World Clock and Day System

The top bar should always expose the world's temporal state: Day 0, simulated calendar date, weekday, 24-hour time, weather, simulation speed, and pause status. Day 0 is initialization. Day 1 is the first complete lived day. The world may start from the real current date for emotional grounding, but simulated time must be independently accelerated.

Support at least four modes. Real-time mode maps one real minute to one simulated minute. Accelerated mode may map one real second to one simulated minute or another configurable ratio. Demo mode compresses a day into a few minutes and prioritizes key events. Pause mode freezes simulation progression while allowing inspection.

The simulation engine owns time. Agents never hallucinate the clock. A tick processor advances by a fixed simulated interval, such as five minutes. It evaluates due schedule transitions, world events, needs, project progress, conversations, and decision triggers. It persists state and emits typed realtime events.

Do not invoke ten agents on every tick. Most ticks should cause zero AI calls. A decision queue identifies only meaningful turns. This creates the illusion of continuous life while keeping model usage sparse and intentional.

6. Creating Multiple Agents from One Grok API

One Grok API can power every agent. You do not need ten keys, ten model deployments, or ten continuously running language models. Agent identity exists in persistent application state and in the context assembled for each invocation. The same model receives different identity, profession, memories, relationships, observations, goals, and allowed actions depending on which agent is taking a turn.

For Mira's turn, the orchestrator loads Mira's identity, current world observations, crop state visible from her location, schedule, goals, recent events, relationship context, and only memories accessible to Mira. For Arun's turn, the same Grok model receives Arun's engineering identity and his own accessible context. The model is shared; the experiential state is not.

Every invocation should use structured output. The model can propose an action, target, spoken text, emotion, user-facing reflection summary, candidate memory, goal update, or hypothesis. Pydantic validates the shape. The action validator checks whether the proposal is legal. The world reducer applies accepted changes.

Concurrency should be limited. A priority queue can serialize most turns and allow a small number of concurrent calls when safe. Urgent emergencies, scheduled high-value conversations, and active discovery chains receive priority. Decorative reflections can be skipped under quota pressure.

6.1 Agent Context Contract

Each AI call should include immutable identity; personality traits; profession and capabilities; current time; location; nearby observable entities; physical state; active goals; today's schedule; trigger event; recent conversation context; relevant scoped memories; public knowledge available to the agent; relationship context; legal action types; and strict output schema.

6.2 Information Isolation

Memory queries must always include owner and scope filters. Private memories belong to one agent. Household memories may be available to household members only if explicitly shared. Public knowledge is globally accessible through appropriate reading or announcements. The system must never retrieve all memories and rely on the prompt to ignore unauthorized ones.

6.3 Cost-Aware Conversation Generation

A two-agent conversation can be generated in one bounded call using both agents' permitted context, then converted into separate listener memories. For critical scenes, separate perspective calls can provide richer individuality. The MVP should use one-call exchanges for cost and latency unless the scene directly supports the core demo.

7. Memory System: The Heart of the Civilization

Memory should not be modeled as a giant transcript. The architecture needs distinct memory types, ownership, provenance, confidence, importance, privacy, and retrieval rules. Supermemory provides semantic long-term retrieval, while Supabase stores structured metadata and world references.

Episodic memory records a lived event. Semantic memory captures generalized knowledge. Social memory records an interaction. Relationship memory captures trust-relevant experience. Procedural memory describes learned methods. Book memory records knowledge obtained through reading. Hypothesis memory is explicitly uncertain. Public knowledge represents validated institutional knowledge.

Every important memory should answer: who owns it, what happened, when, where, how the owner learned it, who the source was, which world event caused it, how confident the owner is, how important it is, whether it is emotionally significant, who may access it, and which later decisions referenced it.

When one agent tells another about an experience, do not copy the original memory verbatim. Create a new derived memory for the listener. If Mira experienced drought and tells Aadi, Aadi remembers that Mira reported dry fields and feared lower harvest. This preserves perspective and provenance.

Retrieval ranking can combine semantic similarity, importance, recency, emotional significance, unresolved-goal relevance, relationship relevance, repetition, and confidence. Old but foundational memories can remain strong through reinforcement. Trivial details naturally fall out of top retrieval without immediate deletion.

7.1 Memory Lifecycle

Event or conversation occurs; candidate experience is extracted; importance and type are classified; duplicates are checked; semantic content is stored in Supermemory; structured metadata is stored in Supabase; later context triggers retrieval; retrieved memory influences action, conversation, or reflection; references are logged for causal traceability.

7.2 Memory Provenance

A provenance graph links original world event to direct memory, transmitted memory, later retrieval, hypothesis, validation, discovery, and consequence. This graph is both an engineering debugging tool and a signature public UI feature.

7.3 Forgetting

The system should model retrieval decay rather than aggressively deleting memories. Relevance can decrease with time unless reinforced by repetition, emotional significance, public recording, or repeated usefulness. This allows a realistic active-memory hierarchy while preserving historical auditability.

7.4 Institutional Memory

Validated discoveries can be documented in the library or town records. This converts private or social experience into durable public knowledge. Future agents can learn it without witnessing the original event, modeling the fundamental civilizational transition from personal memory to culture.

8. The Child Agent and Emergent Intelligence

The child is not simply another role. It is the mechanism that gives the world an intellectual trajectory. Adults spend much of their time specializing. The child crosses boundaries. It asks the farmer about soil, the doctor about illness, the engineer about machines, the artist about stories, and the teacher about books. Over time, its memory graph becomes broad.

Aadi should have a curiosity budget. Each day the child can pursue only a limited number of questions, visits, and deep reflections. Scarcity makes choices meaningful and controls API cost. The child's next visit can be selected from unresolved questions, recent emotional experiences, novel events, trusted relationships, and semantically connected memories.

Nightly reflection retrieves a small set of relevant experiences. Grok is asked to identify unresolved questions and possible connections. A candidate connection becomes a hypothesis, not fact. The child can create a goal to ask an expert, read a book, observe a phenomenon, or perform a bounded experiment.

Growth should be measured through breadth, depth, synthesis, and validation. Suggested metrics include concepts encountered, unique domains visited, trusted expert relationships, questions resolved, cross-domain links, hypotheses created, hypotheses validated, discoveries adopted, and civilization impact.

The child must be allowed to fail. Some hypotheses should be rejected. This makes successful discoveries more credible and creates learning experiences. A rejected idea can itself become a memory that influences future reasoning.

8.1 Irrigation Discovery Example

Day 4: Mira directly experiences dry soil and falling crop health. Evening: Mira tells a household member. Day 5: Aadi hears the report and stores a derived memory. Day 6: During a workshop visit, Arun explains that channels and pipes can move water. Night reflection retrieves both memories. Aadi asks whether water can be brought from the reservoir to the field. The system creates an unvalidated hypothesis. Arun evaluates slope, materials, labor, and known technology. If feasible, the leader receives a project proposal. Construction consumes resources over several days. Crop moisture improves. The provenance view displays the entire chain.

8.2 Creativity Without Magic

A candidate discovery should cite supporting memories and respect world constraints. Novelty can be approximated by cross-domain distance; usefulness by expert validation and problem relevance; feasibility by deterministic checks. The LLM never grants itself impossible technology.

9. Deterministic World Engine

The LLM should not own truth. The deterministic world engine is authoritative for time, location, resources, inventories, weather, crop status, health, buildings, schedules, projects, and valid actions. Grok proposes behavior inside this world. Application code validates and applies it.

This separation prevents teleportation, impossible construction, invented resources, and knowledge leakage. If an agent is at the farm, a private face-to-face conversation with someone at home cannot occur. If only three wood units exist, a twenty-unit building cannot be completed. If an agent never learned a fact, the model should not receive it.

The first map should remain compact: households, farm, workshop, clinic, school/library, market or kitchen, town hall, research garden, central square, reservoir, and paths. A stylized 2D view is enough. The goal is legibility and emotional life, not graphical simulation complexity.

9.1 Resources

Begin with food, water, wood, tools, medicine, books or records, and perhaps simple trade credits. Every resource should create meaningful dependencies. Avoid adding dozens of crafting materials before the core memory loop works.

9.2 Events

Events can be scheduled, probabilistic, state-triggered, or AI-proposed and system-validated. Useful events include rain, drought, crop disease, broken tools, injury, food shortage, contaminated water, festival, discovered plant, interpersonal disagreement, new book, or construction milestone.

9.3 Needs

Use a small need model: energy, hunger, social connection, safety, curiosity, and purpose. Needs influence scheduling and decisions but should not dominate the product. The project is about memory and knowledge, not recreating every mechanic of a life simulator.

10. Low-Level System Architecture

The recommended stack is React, Vite, and TypeScript on the frontend; Python and FastAPI on the backend; LangChain for AI orchestration; Grok API for reasoning and language generation; Supermemory for semantic long-term memory; Supabase PostgreSQL for structured persistence; and WebSocket or Server-Sent Events for live updates.

For the hackathon, use a modular monolith. One FastAPI application can contain logically separated simulation, agent, AI, memory, world, persistence, and realtime modules. Do not begin with Kubernetes, Kafka, ten agent services, or distributed queues. Complexity should be earned by scale, not assumed.

The core flow is: scheduler advances time; tick processor evaluates routines and events; decision queue identifies meaningful turns; context builder reads world state and retrieves scoped memories; LangChain invokes Grok; parser validates structured output; action validator checks legality; reducer updates deterministic state; memory service stores new experiences; event store records consequences; realtime broadcaster pushes updates to React.

10.1 Architecture Diagram

OBSERVER → React/Vite UI → REST + WebSocket/SSE → FastAPI API Layer → Simulation Clock and Scheduler → Tick Processor → Routine Engine + Event Engine → Agent Decision Queue → Context Builder → LangChain Orchestrator → Grok API. Context Builder reads Supabase world truth and retrieves agent-scoped memories from Supermemory. Grok output returns through Pydantic Parser → Action Validator → World Reducer → Supabase persistence → Memory Service → Supermemory → Event Broadcaster → UI.

10.2 Backend Modules

api: routers for world, agents, timeline, memories, discoveries, simulation admin, and health. simulation: clock, scheduler, tick processor, routine engine, event engine. agents: registry, context builder, decision queue, action validator, goal service. ai: orchestrator, chains, prompts, parsers, Grok client. memory: service, retriever, scorer, provenance, transmission. world: map, locations, resources, weather, inventory, crops, health, projects. persistence: Supabase repositories. realtime: broadcaster and event schemas.

10.3 Dependency Direction

API routers do not directly call Grok. The simulation layer decides whether cognition is needed. The agent context builder assembles legal knowledge. The AI layer proposes structured output. Validation protects world truth. Repositories persist state. Memory services enforce ownership and provenance. This direction reduces coupling and prevents accidental bypass of rules.

11. Complete Tick-by-Tick Workflow

At each tick, the clock service computes the next simulated time. The scheduler queries due routine blocks. Deterministic transitions are applied immediately. The event engine checks weather, resource thresholds, project progress, health triggers, and seeded scenario conditions. New observations are created only for agents physically able to perceive them.

Suppose time reaches 08:00. Mira's schedule says farm work. Her movement from home to farm is deterministic. The backend emits AGENT_MOVED and ACTIVITY_CHANGED. No Grok call occurs. At 10:20 crop moisture falls below a threshold. Because Mira is present, the world emits DROUGHT_OBSERVED. A candidate episodic memory is stored.

The decision queue determines that the event is meaningful enough for a bounded agent turn. Mira's context includes dry soil, crop health, weather history available to her, relevant farming memories, and legal actions. Grok may propose inspecting another plot, continuing work, planning to tell the leader, or raising concern at dinner. The validator checks the action.

At dinner, the social scheduler chooses salient memories. A conversation call receives the participants' legal context. Dialogue is generated. Aadi or another listener receives a derived memory with source attribution. The original farmer memory remains private unless explicitly shared.

At nightly reflection, Aadi's unresolved concern about food triggers semantic retrieval. The drought memory and a prior engineering explanation are retrieved. Grok produces a hypothesis. The parser marks it unvalidated. A future goal is created to ask Arun. The next day's flexible schedule gives that visit higher priority.

During the expert consultation, Arun receives the proposal and relevant engineering facts, but not unrelated private memories. He can validate, reject, or request more information. If validated, a project proposal enters leadership workflow. Rohan can approve it based on resources and priorities. Construction then progresses deterministically.

Every stage writes an event record. The UI can reconstruct the causal sequence without relying on hidden chain-of-thought.

12. API Design

REST endpoints provide snapshots, history, and inspection. Realtime channels provide ongoing events. Suggested endpoints include GET /api/v1/world, GET /world/time, GET /world/map, GET /agents, GET /agents/{id}, GET /agents/{id}/schedule, GET /agents/{id}/relationships, GET /agents/{id}/memories, GET /timeline, GET /discoveries, GET /discoveries/{id}, and GET /projects.

Admin or demo endpoints include POST /admin/simulation/pause, /resume, /speed, /reset, and optionally /seed-event. These must be protected. A public observer should not accidentally alter the world.

Realtime event types include TIME_UPDATED, WEATHER_CHANGED, AGENT_MOVED, ACTIVITY_CHANGED, CONVERSATION_STARTED, MESSAGE_SPOKEN, CONVERSATION_ENDED, MEMORY_CREATED, MEMORY_TRANSMITTED, GOAL_CREATED, HYPOTHESIS_CREATED, HYPOTHESIS_VALIDATED, HYPOTHESIS_REJECTED, DISCOVERY_RECORDED, PROJECT_STARTED, PROJECT_PROGRESS, PROJECT_COMPLETED, RESOURCE_CHANGED, and RELATIONSHIP_CHANGED.

Each event should include event_id, simulation_time, day_number, event_type, actor IDs, location, public payload, and correlation ID. Sensitive memory content should not be broadcast unless the observer is allowed to see it by product design.

13. Database Design

Supabase PostgreSQL should be the structured source of truth. Core tables include simulation_runs, world_clock, agents, households, relationships, locations, buildings, schedules, routine_blocks, agent_state, world_resources, inventories, weather_state, crop_state, health_state, world_events, conversations, messages, memory_metadata, goals, hypotheses, discoveries, discovery_sources, projects, project_tasks, books, public_knowledge, personality_traits, needs, and AI_turn_logs.

The agents table stores stable identity. Agent state stores rapidly changing status. Relationships are directional because trust and closeness may differ between two people. Memory metadata references the external Supermemory object and stores ownership, type, importance, confidence, provenance, privacy, and timestamps.

World events form the audit backbone. Every consequential state change should be reconstructable from events. AI turn logs record trigger, agent, retrieved memory IDs, prompt version, model latency, output status, proposed action, accepted action, rejection reason, and token information when available.

13.1 Suggested Agent Fields

id, simulation_run_id, name, age, age_stage, role, household_id, baseline_personality, capability_profile, communication_style, created_at, updated_at.

13.2 Suggested Memory Metadata Fields

id, external_memory_id, owner_agent_id, memory_type, summary, importance, emotional_weight, confidence, source_type, source_agent_id, source_event_id, source_memory_id, location_id, privacy_scope, occurred_at, created_at, last_retrieved_at, reinforcement_count.

13.3 Suggested Discovery Fields

id, creator_agent_id, title, description, hypothesis_id, validation_status, validator_agent_id, created_at, validated_at, civilization_impact, public_knowledge_id. A separate discovery_sources table links all supporting memory IDs.

14. Frontend UI/UX Design

The UI must make the world understandable within seconds. The default screen is a stylized 2D settlement. The top bar shows Day N, date, 24-hour clock, weather, speed, and pause status. The left side lists agents with portraits, profession, current activity, and small status indicators. The center contains the world map. The right side becomes an inspector for the selected agent or event. The bottom can hold a timeline or event ticker.

Agent avatars move between buildings along paths. A selected agent receives a clear highlight and optional follow-camera behavior. Brief speech bubbles appear during conversations. Important events receive restrained icons: memory, discovery, illness, conflict, milestone, project, or public announcement.

The observer can switch among agents like following NPCs in a game. The agent inspector should show current activity, location, destination, mood, active goal, next schedule item, today's experiences, important long-term memories, relationships, known domains, and hypotheses. Tabs prevent information overload.

Do not display hidden chain-of-thought. Show concise public-facing reflections such as “I am worried the dry field may reduce our harvest” and provenance such as “Experienced directly on Day 4” or “Heard from Mira on Day 5.”

14.1 Main World Screen

Top: world clock and controls. Left: agent roster. Center: map. Right: context inspector. Bottom: event timeline. Filters can isolate one agent, household, profession, conversation, memory event, discovery, or project.

14.2 Follow Mode

Clicking an agent enters follow mode. The camera centers or highlights that avatar. The UI shows a live activity feed: “08:02 walking to workshop,” “08:15 repairing plow,” “09:40 speaking with Mira,” “09:47 formed memory.” The user can jump instantly to another agent.

14.3 Memory Cards

Memory cards use distinct source icons: eye for direct observation, speech for told by another agent, book for reading, spark for hypothesis, monument or archive for public knowledge. Each card includes day, source, confidence, importance, and links to later consequences when relevant.

14.4 Discovery Provenance Graph

This is the signature screen. Nodes represent original events, memories, conversations, retrieved concepts, hypothesis, expert validation, project, and consequence. Edges show causal transmission. The irrigation demo can visually prove that no single agent originally possessed the full solution.

14.5 Art Direction

Use a warm illustrated diorama or clean 2D game aesthetic. Buildings should be recognizable. Avatars need distinct silhouettes and profession cues. Day/night lighting can change gradually. Avoid photorealistic 3D for the MVP because it consumes development time without strengthening the memory thesis.

15. Frontend Technical Architecture

Use React, Vite, and TypeScript. For ten agents, a DOM, SVG, or lightweight canvas map is sufficient. PixiJS can be introduced if richer sprite rendering becomes necessary, but it should not be mandatory for the first vertical slice.

Server truth remains on the backend. The frontend receives a world snapshot at load and incremental realtime events afterward. Zustand or another small client store can track selected agent, camera mode, filters, local animation state, connection state, and recent events.

Suggested components include AppShell, WorldTopBar, WorldClock, WeatherBadge, SpeedControl, AgentRoster, AgentListItem, WorldMap, Building, Path, AgentSprite, SpeechBubble, FollowCamera, AgentInspector, ScheduleTab, MemoryTab, RelationshipTab, GoalTab, EventTimeline, MemoryCard, DiscoveryGraph, ProjectPanel, ConnectionStatus, and DemoControls.

Typed event reducers update frontend state. Animation should be cosmetic and derived from backend truth. If the backend says an agent moved from home to farm at a given simulated time, the frontend interpolates movement but never independently changes canonical location.

16. Backend File and Folder Structure

A recommended monorepo contains frontend/ and backend/. Under backend/app: main.py; api/routers/world.py, agents.py, memories.py, timeline.py, discoveries.py, projects.py, simulation_admin.py, health.py; core/config.py, logging.py, security.py; simulation/clock.py, scheduler.py, tick_processor.py, routine_engine.py, event_engine.py, scenario_engine.py; agents/registry.py, context_builder.py, decision_queue.py, action_validator.py, goal_service.py; ai/orchestrator.py, grok_client.py, parsers.py, chains/, prompts/; memory/service.py, retriever.py, scorer.py, provenance.py, transmission.py; world/map.py, locations.py, resources.py, weather.py, crops.py, health.py, inventory.py, projects.py; services/conversation_service.py, relationship_service.py, discovery_service.py, book_service.py; repositories/; realtime/broadcaster.py; schemas/; tests/.

The frontend can contain src/components/world, agents, memory, discoveries, timeline, layout; hooks/useWorldStream.ts and useSelectedAgent.ts; services/api.ts and realtime.ts; stores/worldStore.ts and uiStore.ts; types/events.ts, agents.ts, memory.ts; assets for map and sprites.

The folder structure should mirror responsibility boundaries. Simulation code should not contain React concerns. Grok integration should not directly update SQL tables. Memory retrieval should not bypass ownership checks. Action validation should be reusable across all agents.

17. LangChain Orchestration

LangChain should be used as orchestration glue, not as unnecessary abstraction. Define focused chains for agent decision, conversation, reflection, hypothesis generation, expert validation, and daily summary. Each chain has a strict prompt contract and Pydantic output schema.

The context builder prepares only relevant information. Avoid sending an agent's entire life history. Retrieve top memories, include current observations, recent events, active goals, and relationship context. Summarize older conversation history. This reduces cost and improves focus.

A decision chain should return a legal action enum, target reference, spoken text if applicable, emotion, user-facing reflection, candidate memories, and goal changes. A reflection chain should return unresolved questions, possible connections, and hypotheses. An expert-validation chain should return feasibility status, reasons grounded in supplied facts, prerequisites, and recommended next action.

Prompt versions should be tracked. If behavior changes, logs must identify which prompt produced it. This is especially important in autonomous simulations where a subtle prompt edit can alter many downstream events.

18. Books, Reading, and Public Knowledge

Books provide knowledge that does not originate in direct experience. The library can begin with a small original corpus covering agriculture, mechanics, health, weather, mathematics, stories, and ethics. Reading is an action constrained by schedule, location, and attention.

A book memory records title, concept, source, time, and confidence. The child may later connect a book concept with a lived experience. A validated discovery can be written into public records, creating civilization-level institutional memory.

This mechanism supports genuine progression. The society can begin with limited public knowledge, discover something through experience, validate it, document it, and make it teachable to future agents. That is a simple but powerful model of cultural accumulation.

19. Families, Relationships, and Social Memory

Home life should create repeated, meaningful communication. Not every adult needs a separate spouse and child because the roster is limited to ten. Design several households, friendships, mentorships, and professional relationships. The child belongs to one household but has connections across the settlement.

Relationships can track trust, closeness, respect, familiarity, conflict, mentorship, and recent interaction. These values influence what agents choose to share, how strongly they trust hearsay, and whom they seek for help.

At dinner, the memory-sharing selector chooses experiences. A spouse may hear work frustration. The child may receive a simplified explanation. A trusted friend may hear a sensitive concern. Public announcements transmit information broadly. Different channels produce different memory confidence and scope.

Repeated retelling can create rumor chains. The MVP need not simulate complex misinformation, but provenance should preserve hops so future versions can model distortion and confidence decay.

20. Emergence, Validation, and World Development

The world should move forward automatically, but autonomy must be bounded. Deterministic rules define what exists and what is possible. Memories and AI reasoning choose among valid possibilities. Consequences alter future state. This is bounded emergence.

A discovery pipeline should require problem signal, supporting memories, candidate hypothesis, domain validation, resource validation, leadership or project approval where needed, implementation, and observed consequence. This prevents unsupported magical inventions.

The child may propose an irrigation channel, but the engineer checks slope and materials. The leader checks priorities. The inventory service checks resources. Construction takes time. The farm model later measures effect. The discovery is meaningful because it changes future experience.

The same architecture supports other chains: doctor notices illness; child remembers stagnant water; engineer discusses drainage; sanitation project reduces disease. Merchant notices tool shortage; researcher identifies stronger material; engineer tests it; production improves. These chains make the civilization feel cumulative.

21. Pros and Cons

The greatest strength is originality. Many hackathon projects are retrieval chatbots. A visible autonomous society where memories persist, travel, and create discoveries is memorable. It demonstrates the sponsor technology through behavior rather than a hidden API call.

The second strength is visual storytelling. Judges can watch an event happen, inspect the memory, follow social transmission, and see a later consequence. This creates a natural demo narrative.

The third strength is architectural depth. The project touches persistent identity, semantic memory, provenance, agent isolation, realtime systems, deterministic simulation, structured LLM output, and bounded emergence. It can become a platform for villages, companies, laboratories, schools, or colonies.

The primary weakness is scope. Building ten agents, a world engine, memory, relationships, AI orchestration, realtime UI, and discovery mechanics is ambitious. The solution is a vertical slice and strict postponement of nonessential systems.

The second weakness is cost. Ten continuously reasoning agents can exhaust free quotas. The solution is deterministic routine, sparse AI turns, retrieval limits, token budgets, batching, and accelerated but event-focused simulation.

The third weakness is unpredictability. A live demo cannot wait for accidental emergence. Seed a meaningful problem and design reliable opportunities for memory transmission while keeping actual reasoning genuine.

The fourth weakness is coherence. Without validation, agents may invent facts or resources. The deterministic world engine and scoped context are mandatory.

The fifth weakness is pitch risk. If described as “AI NPCs,” the project sounds like entertainment only. The pitch should emphasize persistent multi-agent memory, social knowledge transfer, cross-domain synthesis, and traceable causality.

22. Free-Tier Strategy and Performance

The most important cost optimization is to avoid using the LLM as a clock. Walking, sleeping, routine work, crop growth, resource transfer, project progress, and schedule transitions should be application logic. Grok is reserved for meaningful cognition.

Use one Grok integration for all agents. Cache immutable identity fragments. Retrieve only top relevant memories. Summarize old conversations. Limit daily reflections. Give every agent a daily AI-turn budget. Reserve capacity for critical discovery chains.

Supabase stores structured truth. Supermemory stores semantic long-term memory. React can deploy on free static hosting. FastAPI can use a free or student-friendly backend tier. Exact quotas change and should be checked immediately before deployment.

The system should degrade gracefully. If a low-priority reflection fails, routine continues. If a conversation call fails, retry with bounded backoff or defer it. If the realtime connection drops, the frontend reconnects and fetches a fresh snapshot. Critical demo state should persist.

Prewarm the backend before judging. Include a reset script and health dashboard. Log latency and model failures. Avoid launching ten simultaneous requests.

23. Security, Safety, and Ethics

API keys must remain server-side. Admin controls must be protected. Memory access must enforce ownership and privacy scopes. Database policies should prevent accidental public mutation if Supabase is directly accessible from the frontend.

The world must be clearly labeled as an artificial simulation, not a scientific predictor of human society. Avoid claims about real psychology, politics, or policy. The project demonstrates artificial-agent behavior under designed rules.

Do not expose hidden chain-of-thought. Show concise reflections, memories, goals, actions, and provenance. This provides explainability without pretending to reveal private internal reasoning.

Because a child agent is present, public content should remain age-appropriate. The child exists to learn, ask questions, and synthesize knowledge. Moderation and scenario constraints should match a public hackathon demo.

24. Testing and Reliability

Unit tests should cover clock advancement, schedule transitions, event triggers, action validation, location constraints, resource consumption, project prerequisites, memory scoring, provenance creation, relationship updates, and child-growth metrics.

Integration tests should execute the complete memory chain: world event occurs; observer agent perceives it; memory is stored; agent communicates it; listener receives a derived memory; child retrieves it later; hypothesis is created; expert validates; project begins; world consequence appears.

Knowledge-boundary tests are essential. Ask an agent about an event it neither witnessed nor learned. The expected answer is uncertainty. Retrieval tests should verify that one agent's private memories never appear in another agent's context.

Scenario tests should use fixed random seeds. Drought, broken tool, illness, shortage, and contaminated-water scenarios can verify coherent behavior. The hackathon demo scenario should be tested repeatedly from reset.

Load tests should verify realtime event handling and ensure a burst of agent movements does not overload the UI. AI calls should have timeouts, retries, and circuit-breaking behavior appropriate to the hosting environment.

25. Observability and Developer Tools

Autonomous systems become impossible to debug without traceability. Every AI turn should have a correlation ID, agent ID, trigger, world snapshot reference, retrieved memory IDs, prompt version, latency, model status, proposed structured action, validation result, and token usage when available.

Create a developer-only turn inspector. It should show why the agent was invoked, what legal observations were supplied, which memories were retrieved, what action was proposed, and whether it was accepted. This will save enormous time.

The public observer UI receives a simplified causal view. It can say that a decision was influenced by a drought memory and an engineering lesson without exposing hidden reasoning.

Daily reports should be generated from structured events, not invented from scratch. The report can summarize major experiences, conversations, discoveries, project progress, and unresolved problems.

26. Hackathon MVP Scope

The MVP should prove the thesis, not simulate all civilization. Show ten named agents on one map. Give each a routine. Implement one accelerated clock. Use deterministic movement. Enable a limited number of meaningful AI decisions and conversations. Store persistent memories. Implement evening transmission. Give the child a reflection loop. Seed one problem. Produce one validated cross-domain discovery. Show provenance.

If development time becomes tight, keep all ten visually present but make five roles deeply active in the core scenario: farmer, engineer, teacher, leader, and child. Other agents can follow deterministic routines and participate lightly. This is better than ten shallow, unreliable AI loops.

The definition of done is a resettable demo where the observer can watch the farmer experience a problem, inspect the resulting memory, watch it reach the child, watch the child acquire a second relevant concept, see a hypothesis form, see expert validation, and see the world change.

26.1 Three-Minute Judge Demo

First 30 seconds: show Day 0 or current day, clock, weather, moving agents, and follow mode. Next 30 seconds: farmer observes drought and a memory is created. Next 30 seconds: evening conversation transfers the experience. Next 30 seconds: child visits engineer and gains a water-transport concept. Next 30 seconds: nightly reflection combines the two memories into an irrigation hypothesis. Final 30 seconds: show expert validation, project creation, and provenance graph. End with the statement that no single agent originally possessed the full solution.

27. Development Roadmap

Phase 1: create monorepo, FastAPI, React, configuration, Supabase schema, health endpoint, and basic deployment. Phase 2: implement simulation clock, world state, locations, ten agents, schedules, and deterministic movement. Phase 3: build map UI, roster, clock, inspector, realtime stream, and timeline.

Phase 4: integrate Grok through LangChain with structured Pydantic output for one agent. Add action validation. Phase 5: integrate Supermemory with agent-scoped storage and retrieval. Phase 6: implement conversations and derived listener memories. Phase 7: implement child reflection, hypotheses, and expert validation. Phase 8: add project consequence and discovery provenance graph.

Phase 9: build seeded demo scenario, reset endpoint, speed controls, logs, retries, and failure fallbacks. Phase 10: polish visuals, onboarding, judge narrative, README, architecture diagram, and recorded backup demo.

28. Future Evolution

After the hackathon, the world can add aging, generations, births and deaths, schools, richer economies, migration, multiple settlements, natural disasters, political institutions, cultural traditions, scientific experimentation, and inter-civilization contact.

A generational system could allow the child to become an adult carrying accumulated experiences while a new child enters the world. Institutional memory would determine whether knowledge survives individual death. This creates powerful questions about education, archives, culture, and forgetting.

A company version could use ten corporate agents and one cross-functional apprentice. A research-lab version could study distributed discovery. A space-colony version could emphasize scarcity and long-term planning. The underlying platform remains the same: persistent specialized agents, scoped memory, social transmission, deterministic world state, and traceable synthesis.

29. Recommended Final Product Identity

The strongest identity is not “a game with AI characters.” It is “a living memory civilization.” The world moves by itself. Every person has a private past. Experience travels through relationships. Knowledge becomes culture. One child grows by connecting the lives of everyone around them.

A possible product name can evoke memory, society, and emergence, but naming should come after the visual and narrative direction is fixed. The pitch should always foreground the visible causal chain and the role of persistent memory.

The final experience should leave the observer with a specific feeling: they are not chatting with an AI; they are looking through a window into a tiny world whose inhabitants remember yesterday.

30. Final Recommendation

Build the civilization, but build it around one undeniable mechanism: memory changes what becomes possible. The nine specialized adults create diverse experiences. The child creates connective intelligence. Grok provides bounded reasoning. LangChain orchestrates context. Supermemory preserves long-term experience. Supabase preserves deterministic truth and provenance. FastAPI runs the simulation. React makes the world observable.

Do not spend the hackathon trying to simulate every minute with AI. Let code handle routine. Let AI handle meaning. Do not let all agents know everything. Let knowledge travel. Do not let inventions appear because a model said so. Require evidence, expertise, resources, and validation.

The best possible demo is a story the system can prove: something happened; someone remembered it; someone else learned it; a different experience became relevant; the child connected them; an expert validated the idea; the civilization changed. That is a complete narrative, a strong technical architecture, and a compelling demonstration of persistent memory.

Appendix A. Extended Engineering Notes

Simulation clock engineering — deep implementation note 1. The clock should be monotonic within a run and persisted frequently enough that a backend restart cannot silently rewind the civilization. Every event uses simulation time and real creation time. Speed changes affect future tick scheduling but never rewrite historical timestamps. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Decision queue design — deep implementation note 1. The queue ranks triggers by urgency, narrative importance, unresolved goals, scheduled obligation, emergency severity, observer relevance, and remaining AI budget. A starvation guard prevents one agent from monopolizing cognition. Routine events bypass the queue. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Context assembly — deep implementation note 1. Context is a security boundary. The builder joins current observations, agent identity, legal world facts, recent events, active goals, relationship data, public knowledge, and scoped memory retrieval. It should explicitly omit inaccessible private information. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Memory scoring — deep implementation note 1. A useful initial rank combines semantic relevance, importance, recency, emotional weight, goal relevance, relationship relevance, confidence, and reinforcement. Keep the formula inspectable. Log why each memory was selected. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Conversation scheduling — deep implementation note 1. Conversations require co-location or an explicit communication channel, compatible availability, social motivation, and topic candidates. The system should avoid endless random chatter. Important experiences and active goals provide topics. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Relationship evolution — deep implementation note 1. Trust and closeness change slowly. One friendly sentence should not transform a relationship. Use bounded deltas, decay, repeated evidence, and event significance. Relationship changes should cite causal interactions. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Hypothesis management — deep implementation note 1. Hypotheses need status values such as proposed, investigating, supported, rejected, validated, and archived. They link supporting and contradicting memories. Rejection should become a learning event. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Project system — deep implementation note 1. Validated ideas become projects only after prerequisites are satisfied. Projects have tasks, required resources, responsible agents, progress, blockers, start time, and expected consequence. The world engine advances work. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Agent personality — deep implementation note 1. Personality should influence preference and style without overriding world constraints. Traits can affect risk tolerance, sociability, curiosity, patience, sharing tendency, and openness to novelty. Keep trait effects bounded. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Books and education — deep implementation note 1. Reading consumes simulated time and creates source-tagged memories. Teaching can transfer concepts at a rate affected by relationship, attention, and prior knowledge. Public records preserve validated knowledge. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality.

The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Realtime frontend — deep implementation note 1. The browser loads a snapshot and then subscribes to events. Sequence numbers detect missed messages. On reconnect, the client requests events after the last known sequence or reloads a snapshot. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Map and movement — deep implementation note 1. Movement should follow simple paths and estimated travel durations. The backend owns departure and arrival truth. The frontend interpolates animation. Location gates observation and conversation. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Weather and agriculture — deep implementation note 1. Weather can be generated from a seeded simple model. Crop state depends on moisture, season, labor, pests, and interventions. The system need not be scientifically exact; it must be internally consistent. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Health system — deep implementation note 1. Health events should remain simple and non-graphic. Symptoms, injury, fatigue, nutrition, and sanitation can create cross-domain problems. The doctor validates health-related proposals within simulation rules. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Discovery provenance — deep implementation note 1. Every discovery should expose source memories, their owners, original events, transmission edges, retrieval moments, hypothesis creator, validator, and consequence. This is the clearest proof of memory utility. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Demo scenario design — deep implementation note 1. Seed the problem but not the answer. Arrange schedules so information can plausibly travel. Guarantee opportunities for the child to encounter complementary knowledge. Preserve genuine model reasoning within bounded choices. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Failure recovery — deep implementation note 1. Persist accepted state before broadcasting consequences. Use idempotency keys for tick processing and AI result application. A retry must not duplicate memories, consume resources twice, or complete a project twice. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Token budgeting — deep implementation note 1. Assign budgets by simulation day, agent, and chain type. Critical validation and synthesis calls receive reserved capacity. Decorative conversations are first to be reduced when quotas tighten. In implementation pass 1, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

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Relationship evolution — deep implementation note 2. Trust and closeness change slowly. One friendly sentence should not transform a relationship. Use bounded deltas, decay, repeated evidence, and event significance. Relationship changes should cite causal interactions. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Hypothesis management — deep implementation note 2. Hypotheses need status values such as proposed, investigating, supported, rejected, validated, and archived. They link supporting and contradicting memories. Rejection should become a learning event. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Project system — deep implementation note 2. Validated ideas become projects only after prerequisites are satisfied. Projects have tasks, required resources, responsible agents, progress, blockers, start time, and expected consequence. The world engine advances work. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Agent personality — deep implementation note 2. Personality should influence preference and style without overriding world constraints. Traits can affect risk tolerance, sociability, curiosity, patience, sharing tendency, and openness to novelty. Keep trait effects bounded. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Books and education — deep implementation note 2. Reading consumes simulated time and creates source-tagged memories. Teaching can transfer concepts at a rate affected by relationship, attention, and prior knowledge. Public records preserve validated knowledge. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Realtime frontend — deep implementation note 2. The browser loads a snapshot and then subscribes to events. Sequence numbers detect missed messages. On reconnect, the client requests events after the last known sequence or reloads a snapshot. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Map and movement — deep implementation note 2. Movement should follow simple paths and estimated travel durations. The backend owns departure and arrival truth. The frontend interpolates animation. Location gates observation and conversation. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Weather and agriculture — deep implementation note 2. Weather can be generated from a seeded simple model. Crop state depends on moisture, season, labor, pests, and interventions. The system need not be scientifically exact; it must be internally consistent. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Health system — deep implementation note 2. Health events should remain simple and non-graphic. Symptoms, injury, fatigue, nutrition, and sanitation can create cross-domain problems. The doctor validates health-related proposals within simulation rules. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Discovery provenance — deep implementation note 2. Every discovery should expose source memories, their owners, original events, transmission edges, retrieval moments, hypothesis creator, validator, and consequence. This is the clearest proof of memory utility. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Demo scenario design — deep implementation note 2. Seed the problem but not the answer. Arrange schedules so information can plausibly travel. Guarantee opportunities for the child to encounter complementary knowledge. Preserve genuine model reasoning within bounded choices. In implementation pass 2, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

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Token budgeting — deep implementation note 3. Assign budgets by simulation day, agent, and chain type. Critical validation and synthesis calls receive reserved capacity. Decorative conversations are first to be reduced when quotas tighten. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Prompt governance — deep implementation note 3. Version every prompt and output schema. Store model name and temperature. Regression-test important scenarios after changes. Autonomous behavior can drift from small prompt edits. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Privacy scopes — deep implementation note 3. Memories can be private, relationship-shared, household, professional, public, or archived. Access is enforced in queries and services, not merely stated in prompts. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

UI explainability — deep implementation note 3. Every major action should expose a concise why: scheduled routine, urgent event, active goal, relationship request, or memory influence. Avoid overwhelming users with raw logs. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Performance — deep implementation note 3. Keep tick processing lightweight. Batch database reads, cache immutable agent identity, use indexed event and memory metadata tables, and avoid full-world prompt serialization. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Testing — deep implementation note 3. Use deterministic seeds and fake model outputs for unit and integration tests. Keep a small suite of golden scenarios. Test forbidden knowledge, impossible actions, duplicate retries, and restart recovery. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Hackathon presentation — deep implementation note 3. Show the world first, architecture second. Let judges witness the causal memory chain before explaining implementation. The visual provenance graph should be the climax. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Post-hackathon scale — deep implementation note 3. Only after the MVP works should the architecture consider durable task queues, distributed workers, multiple simulation shards, larger agent populations, and richer rendering. In implementation pass 3, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Simulation clock engineering — deep implementation note 4. The clock should be monotonic within a run and persisted frequently enough that a backend restart cannot silently rewind the civilization. Every event uses simulation time and real creation time. Speed changes affect future tick scheduling but never rewrite historical timestamps. In implementation pass 4, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Decision queue design — deep implementation note 4. The queue ranks triggers by urgency, narrative importance, unresolved goals, scheduled obligation, emergency severity, observer relevance, and remaining AI budget. A starvation guard prevents one agent from monopolizing cognition. Routine events bypass the queue. In implementation pass 4, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Context assembly — deep implementation note 4. Context is a security boundary. The builder joins current observations, agent identity, legal world facts, recent events, active goals, relationship data, public knowledge, and scoped memory retrieval. It should explicitly omit inaccessible private information. In implementation pass 4, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

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Simulation clock engineering — deep implementation note 5. The clock should be monotonic within a run and persisted frequently enough that a backend restart cannot silently rewind the civilization. Every event uses simulation time and real creation time. Speed changes affect future tick scheduling but never rewrite historical timestamps. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Decision queue design — deep implementation note 5. The queue ranks triggers by urgency, narrative importance, unresolved goals, scheduled obligation, emergency severity, observer relevance, and remaining AI budget. A starvation guard prevents one agent from monopolizing cognition. Routine events bypass the queue. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Context assembly — deep implementation note 5. Context is a security boundary. The builder joins current observations, agent identity, legal world facts, recent events, active goals, relationship data, public knowledge, and scoped memory retrieval. It should explicitly omit inaccessible private information. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Memory scoring — deep implementation note 5. A useful initial rank combines semantic relevance, importance, recency, emotional weight, goal relevance, relationship relevance, confidence, and reinforcement. Keep the formula inspectable. Log why each memory was selected. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Conversation scheduling — deep implementation note 5. Conversations require co-location or an explicit communication channel, compatible availability, social motivation, and topic candidates. The system should avoid endless random chatter. Important experiences and active goals provide topics. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Relationship evolution — deep implementation note 5. Trust and closeness change slowly. One friendly sentence should not transform a relationship. Use bounded deltas, decay, repeated evidence, and event significance. Relationship changes should cite causal interactions. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Hypothesis management — deep implementation note 5. Hypotheses need status values such as proposed, investigating, supported, rejected, validated, and archived. They link supporting and contradicting memories. Rejection should become a learning event. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Project system — deep implementation note 5. Validated ideas become projects only after prerequisites are satisfied. Projects have tasks, required resources, responsible agents, progress, blockers, start time, and expected consequence. The world engine advances work. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Agent personality — deep implementation note 5. Personality should influence preference and style without overriding world constraints. Traits can affect risk tolerance, sociability, curiosity, patience, sharing tendency, and openness to novelty. Keep trait effects bounded. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Books and education — deep implementation note 5. Reading consumes simulated time and creates source-tagged memories. Teaching can transfer concepts at a rate affected by relationship, attention, and prior knowledge. Public records preserve validated knowledge. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Realtime frontend — deep implementation note 5. The browser loads a snapshot and then subscribes to events. Sequence numbers detect missed messages. On reconnect, the client requests events after the last known sequence or reloads a snapshot. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Map and movement — deep implementation note 5. Movement should follow simple paths and estimated travel durations. The backend owns departure and arrival truth. The frontend interpolates animation. Location gates observation and conversation. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Weather and agriculture — deep implementation note 5. Weather can be generated from a seeded simple model. Crop state depends on moisture, season, labor, pests, and interventions. The system need not be scientifically exact; it must be internally consistent. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Health system — deep implementation note 5. Health events should remain simple and non-graphic. Symptoms, injury, fatigue, nutrition, and sanitation can create cross-domain problems. The doctor validates health-related proposals within simulation rules. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

Discovery provenance — deep implementation note 5. Every discovery should expose source memories, their owners, original events, transmission edges, retrieval moments, hypothesis creator, validator, and consequence. This is the clearest proof of memory utility. In implementation pass 5, the team should verify four invariants: the behavior remains grounded in authoritative world state; agent-specific knowledge boundaries are preserved; the event is observable and reconstructable through logs and provenance; and the mechanism remains affordable under a limited Grok quota. Important transitions should use explicit schemas rather than unconstrained prose. Every consequential mutation should be idempotent where retries are possible. When a memory influences a later action, record the reference so the public provenance view and developer debugger can reconstruct causality. The design should prefer a simple rule that can be tested over a sophisticated rule that cannot be explained. If the mechanism does not strengthen persistence, social transmission, discovery, coherence, or the observer experience, it should probably be postponed until after the hackathon.

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